A REPORT ON THE LIMITS OF FURTHER STRENGTH

The Ceiling

Why the current computer is already hard to improve — and why more search, more training, or a bigger table does not change that.

EXACT EXPECTED SCORE

254.143784

points per game under this repo's rules

The first eleven rounds are solved by backward induction over a 786,432-state table. Every hold and every reroll is enumerated. No policy can average more.

01 **Expected score is closed.**
The table is the Bellman fixed point, not a heuristic. Self-play lands on it: 254.16 over 5,000 games.

02 **Match play is the only legal headroom.**
The exact Match-W endgame adds 0.425 percentage points at round 12. That is about one extra win per 235 games.

03 **The full match solution does not scale.**
Two cards multiply the state space to roughly 6×10^{11} before score totals, turn, and round.

04 **The remaining gains are hard to prove.**
A 100-game sample has a 95% interval about 20 points wide. It cannot resolve a 0.4-point effect.

BOTTOM LINE

The computer can be made more annoying. It cannot be made much stronger in a fair game without modelling the opponent or changing the rules.

Prepared from the committed engine, generated table, 5,000-game self-play run, and 100-game CPU benchmark.

No code was changed for this report. Figures are rounded for layout; exact values appear in the cited reports.

01 The expected-score problem is closed

The table is not an approximation of good play. It is the solution of the game under these rules.

The engine's state is `(B, u, y)`: the set of open boxes, the upper-section total capped at 63, and a flag for whether the Yahtzee box already holds 50. For every state, `W(B,u,y)` is the expected score from the start of a turn with optimal play.

$$W(B,u,y) = \Sigma_s\, P(s) \cdot \max_{c \in B} [\, \text{score}_c(s) + \text{bonus}(s) + W(B \setminus c,\, u',\, y') \,]$$

The maximum is over every open box; bonus(s) is the +100 extra-Yahtzee award. The future value is already known for one fewer box, so the table is built by backward induction over all 786,432 reachable states.

1. **Every turn is enumerated.**
   There are 252 distinct five-dice states, at most 32 holds, and 7,776 ordered rolls. The turn solver evaluates all of them.

2. **Every future state is in the table.**
   A deviation can only lower the expected sum at the first state where it differs. That is the optimality argument, not a benchmark.

3. **The numbers agree.**
   5,000 seeded self-play games average `254.16` (SD 58.84, 95% CI ±1.63) against the table's `254.143784`.

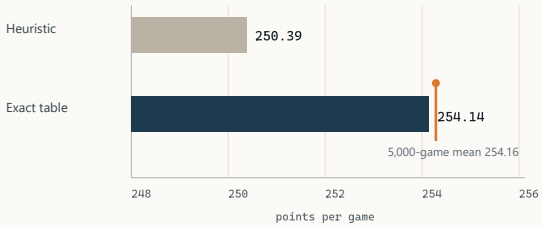
THE TRAINING TRAP

A neural network trained to predict score would be fitting a function the table already knows exactly. It could only approximate it, and an approximation cannot beat the fixed point on average.

Independent rules check: the generator's `--official` mode reproduces the published 254.5896 benchmark within 0.0005. The difference from 254.143784 is this repo's permissive extra-Yahtzee rule, not a solver error.

AVERAGE SCORE PER GAME

5,000 seeded games, same trainer



The exact value is a line, not a bar: 254.143784. The self-play result is a validation.

Heuristic fallback: 250.39. Exact table: 254.143784. The 3.75-point gap is the value of solving the rest of the card exactly; it is already captured.

786k	5,000	±1.63
reachable states in the exact table	seeded self-play games used to validate it	95% confidence interval on the measured mean

02 Match play is the only legitimate headroom — and it is thin

Exact-W maximizes points. Match-W maximizes the chance of beating a particular opponent. The current CPU switches only for the last two rounds.

CURRENT BROWSER POLICY



Rounds 1–11 ignore the scoreboard. Rounds 12–13 are exact match-equity search. Two rounds out of thirteen is the entire shipped match-aware surface.

Why the endgame is tractable. At round 12 the human has exactly one box left, so their category choice is forced. Only their hold policy must be modelled. The CPU's own turn is still solved by exact dice enumeration, and the human's future turn is represented by an exact distribution over the shipped Exact-W policy.

Why the gain is small. On 50 real round-12 states, Match-W was at least as good on every state: 40 improved, 10 tied, none got worse. The average gain was `+0.425 pp`. At round 13 it was `+0.175 pp`. Those are exact probabilities, not simulated win rates.

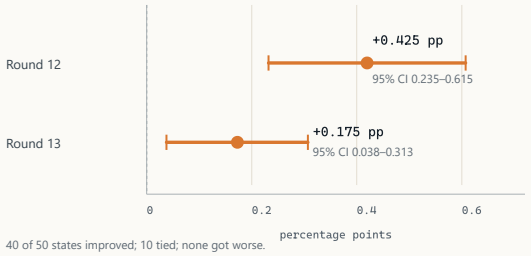
STATE	EXACT-W	MATCH-W	GAIN
Start of round 12	47.1275%	47.5525%	+0.425 pp
Start of round 13	50.6206%	50.7958%	+0.175 pp

THE BENCHMARK REALITY

The 100-game CPU-role run finished **51–49** against a **50–50** Exact-W control. That is one game. The 95% interval on the 51% result is roughly 41–61%. It is not evidence of a large edge.

EXACT GAIN OVER EXACT-W

Percentage points, same states



40 of 50 states improved; 10 tied; none got worse.

Exact state-level evaluation. The horizontal line is the 95% confidence interval; the dot is the mean gain. The absolute effect is small because only the final two rounds are match-aware.

SCALE CHECK

A 0.425-point edge is about **one extra win per 235 games** against an Exact-W opponent. You will not feel that in a five-game session.

03 Exact match play does not scale

The current solver is exact because it starts at round 12. From round 1, both full cards and both totals become part of the state.

The one-card table has **786,432 reachable states** in a 1,048,576-entry array. A full match state needs two of those cards, plus both current totals, whose turn it is, and the round. Even before the base-score totals, the card-pair product is:

$$786{,}432 \times 786{,}432 \approx 6.2 \times 10^{11}$$

card-pair states before score totals, turn, and round. The real count is larger.

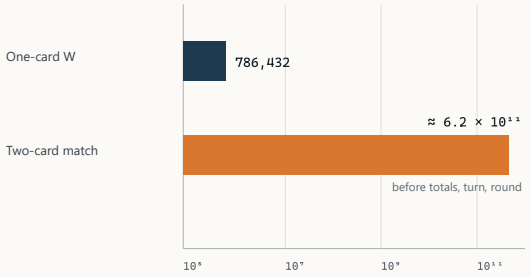
More search is not the missing ingredient. The turn solver already enumerates every legal hold and every reroll outcome. When the continuation values are exact, a deeper search adds no information. The only way to improve is to change the *value* at the end of the turn — from points to match equity — and then the state space changes.

The exact round-11 pilot proves the trade. Starting Match-W one round earlier raised the exact average gain from `+0.425 pp` to `+1.318 pp` in a five-game pilot. It took **349 seconds for five games**, about 70 seconds per game. A Web Worker would stop the interface freezing; it would not stop the wait.

THE PRACTICAL WALL

A full exact table is not a bigger download. It is a different computational universe. The browser can afford the one-card table; it cannot afford the two-card match table.

REACHABLE STATES
logarithmic scale



The ratio is about 786,000× before the extra match dimensions.

Log scale. The apparent "small" step from one card to two is eight orders of magnitude, not a factor of two. The exact two-card table is not a browser artifact.

2
rounds currently solved with match equity

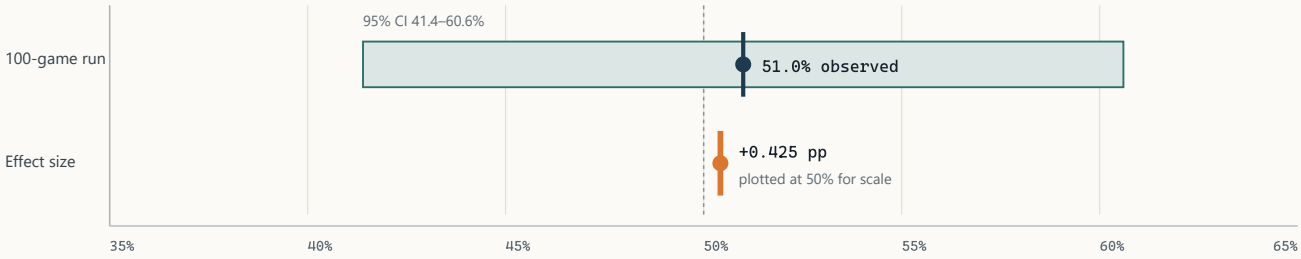
70s
per game for the exact round-11 pilot

04 The remaining gains are smaller than the noise

A 100-game record is a descriptive sample, not a microscope. The effect we are trying to see is about one game in 235.

THE BENCHMARK CANNOT RESOLVE THE EFFECT

100-game win rate vs the size of the exact gain



The interval is about 20 points wide. The gain is 0.425 points. On this scale it is a hairline.

The 100-game estimate is the wide teal interval. The orange mark plots the exact gain at the 50% reference so its width is directly comparable; the actual round-12 rate is 47.55%, not 50%. The benchmark's resolution is much larger than the effect.

What would it take to see it? For a simple 50/50 comparison at 80% power and two-sided $\alpha = 0.05$, the approximate sample sizes are:

TRUE GAIN	GAMES	WHAT IT ESTABLISHES
+0.425 pp	≈ 109,000	the current round-12 effect
+1.3 pp	≈ 11,600	the round-11 pilot effect
+2 pp	≈ 4,900	a small but noticeable edge
+5 pp	≈ 790	a large, easily felt edge

Normal approximation, one-sample win rate against 50%, no pairing. Exact state-level evaluation avoids this cost by evaluating the policy directly.

Why a "better" result can be an illusion. The treatment and control are not paired after round 12: a different hold consumes a different number of dice, so the rest of the game sees a different random stream. A model can also overfit to one seed, or to the Exact-W opponent, and look stronger without being stronger against a human.

> **THE RIGHT TEST**
>
> Use paired states and exact policy evaluation for sub-point effects. Use thousands of games only for effects around 1 point or larger. A 100-game record is a story, not a measurement.

The 51–49 result is one game above the 50–50 control. The 95% interval on 51% is roughly 41–61%; it contains both "no edge" and "a large edge."

05 What could still move it — and what would not

The current CPU is not one clever update away from being much stronger. The remaining levers change the objective, the opponent model, or the rules.

LEVER	WHAT IT CHANGES	HONEST CEILING
Exact Match-W in round 11 exact	Adds one more round of exact match-equity search on both cards.	**+1.3 pp** in a five-game pilot; about **70 s/game**. Not browser-viable as-is.
Approximate all-game match value model	Learns or estimates win probability from both cards, totals, turn, and round; makes the mid-game scoreboard-aware.	Unknown. Can be worse than Exact-W if the value model is wrong. Needs paired validation, not a 100-game story.
Opponent model adaptive	Stops assuming the human plays Exact-W. Learns their holds and banks and chooses a best response.	The largest practical gain against a specific person, but not a universal strength increase. It can be exploited by a player who adapts back.
P(win) instead of P(win) + ½P(tie) objective	Treats ties as losses rather than half-wins; may take more risk.	Tiny, and possibly negative for "unbeaten" rate. A tie is still a tie.
More training on score not useful	Fits a model to predict expected score from the state.	Cannot exceed the exact table. It can only approximate 254.143784 less accurately.
Extra roll, loaded dice, score boost unfair	Changes the game rather than the policy.	Arbitrarily hard, but not smarter. This is difficulty, not strength.

CONCLUSION

Improving this CPU is unlikely because the easiest axis is exhausted: the table is exact, the turn search is exhaustive, and the only exact match-aware extension is both slow and small. What remains is not a better solver. It is a better model of the opponent, an approximation with a new failure mode, or a different game.

THREE THINGS THAT WOULD CHANGE THE ANSWER

1. The goal becomes beating one specific human. Then opponent modelling has real room: the CPU can exploit that player's actual choices, not an imagined Exact-W clone.

2. The rules change. A different Yahtzee variant has a different exact table. The current 254.143784 is a property of these rules, not a universal constant.

3. Unfair difficulty is allowed. An extra roll or loaded dice can make the CPU arbitrarily hard. That is a game-design choice, not an improvement in play.

SOURCES

1. README: exact table, 5,000-game validation, Match-W
2. Match-W design and exact state-level evaluation
3. 100-game CPU-role head-to-head benchmark
4. tools/solve-optimal.js: table generator
5. src/engine.js: exact turn solver
6. src/match-solver.js: browser Match-W endgame
7. tools/test-engine.js: 61 engine checks